

Final Report: Epe1 Histone Demethylase Activity (GO:0032452) — Function Assignment Evaluation

Executive Judgment

Verdict: Over-annotated. The IBA annotation assigning histone demethylase activity (GO:0032452) to *S. pombe* Epe1 is not supported by direct experimental evidence and should be removed or replaced. Four independent lines of evidence converge on this conclusion: (1) the JmjC domain of Epe1 lacks the third histidine of the Fe(II)-binding facial triad that is universally required for catalytic demethylation; (2) no in vitro demethylase activity has ever been detected for Epe1; (3) PomBase curators have placed an explicit NOT|IDA annotation on histone H3 demethylase activity (GO:0141052); and (4) the most mechanistically detailed study to date (Raiymbek et al. 2020) demonstrated that Epe1 opposes heterochromatin assembly through a non-enzymatic protein-protein interaction rather than catalytic demethylation. The current IBA annotation originates from phylogenetic inference (GO_REF:0000033, PANTHER family propagation) and incorrectly transfers demethylase activity from catalytically competent JmjC family members to this divergent, catalytically inactive member.

Summary

The seed hypothesis proposes that *Schizosaccharomyces pombe* Epe1 (UniProt O94603) possesses histone demethylase activity (GO:0032452), based on an IBA (Inferred from Biological Ancestor) annotation propagated through phylogenetic analysis by the GO Consortium (GO_REF:0000033). This report evaluates whether Epe1 directly catalyzes histone demethylation, drawing on primary literature, computational sequence analysis, and publicly available structural data.

The weight of evidence strongly indicates that this annotation is over-annotated. While Epe1 contains a JmjC domain — the signature fold of the 2-oxoglutarate/Fe(II)-dependent dioxygenase superfamily — its catalytic site is degenerate. Computational analysis of the Epe1

sequence reveals that the Fe(II)-binding facial triad, which in active JmjC demethylases consists of an H-x-D/E...H motif, is incomplete: the third histidine is replaced by isoleucine (I379). This substitution is expected to abolish Fe(II) coordination and thereby prevent catalytic turnover. Consistent with this structural prediction, no laboratory has detected histone demethylase activity for Epe1 *in vitro*, despite multiple attempts by groups studying JmjC family enzymes. PomBase curators have formally recorded a NOT annotation (NOT|enables GO:0141052, histone H3 demethylase activity, evidence IDA) referencing the foundational JmjC demethylase paper by Tsukada et al. (2006).

Instead of functioning as a demethylase, Epe1 appears to oppose heterochromatin through a non-enzymatic mechanism involving its JmjC domain as a protein-interaction interface with Swi6/HP1. This model, established by Raiymbek et al. (2020), explains why JmjC domain mutations cause loss of Epe1 function *in vivo* — not because they ablate catalytic activity, but because they disrupt a critical protein-protein interaction regulated by H3K9 methylation. The phylogenetic inference underlying the IBA annotation fails to account for this functional divergence within the JmjC family.

Key Findings

Finding 1: Epe1 JmjC Domain Lacks the Third Histidine of the Fe(II) Facial Triad

The catalytic mechanism of JmjC-domain histone demethylases requires a facial triad of three residues — typically His-x-Asp/Glu...His — that coordinate the Fe(II) cofactor essential for oxidative demethylation. Computational analysis of the Epe1 JmjC domain (residues 243–402) demonstrates that while Epe1 retains the first two elements of this triad (H297 and E299, forming an HxE motif) and the α -ketoglutarate (α -KG) binding residues (T294, K314), the critical third histidine is absent. At the structurally equivalent position (~residue 379), Epe1 has an isoleucine rather than histidine.

This finding was validated against characterized JmjC demethylases:

Enzyme	Triad residues	Third ligand	Catalytically active?
KDM2A/JHDM1	H212, D214, H284	His	Yes
KDM3A/JHDM2A	H1120, D1122, H1249	His	Yes
ScJhd1	H305, D307, H377	His	Yes
FIH (hydroxylase)	H199, D201, H279	His	Yes
Epe1	H297, E299, I379	Ile	No
PHF2	H249, D251, Y321	Tyr	No (in vitro)

UniProt annotates only 4 metal/cofactor binding sites for Epe1 (T294, H297, E299, K314) compared to 5 for catalytically active KDM2A (T209, H212, D214, K229, H284), confirming the missing third coordination site. Notably, the crystal structure of human PHF2 — which shares significant sequence homology with Epe1 including a tyrosine at the corresponding third-ligand position — also shows no in vitro demethylase activity ([PMID: 21167174](#)). The authors of that study explicitly noted: *"PHF2 shares significant sequence homology throughout the entire region, including the above-mentioned tyrosine at the corresponding iron-binding position, with that of Schizosaccharomyces pombe Epe1, which plays an essential role in heterochromatin function but has no known enzymatic activity."*

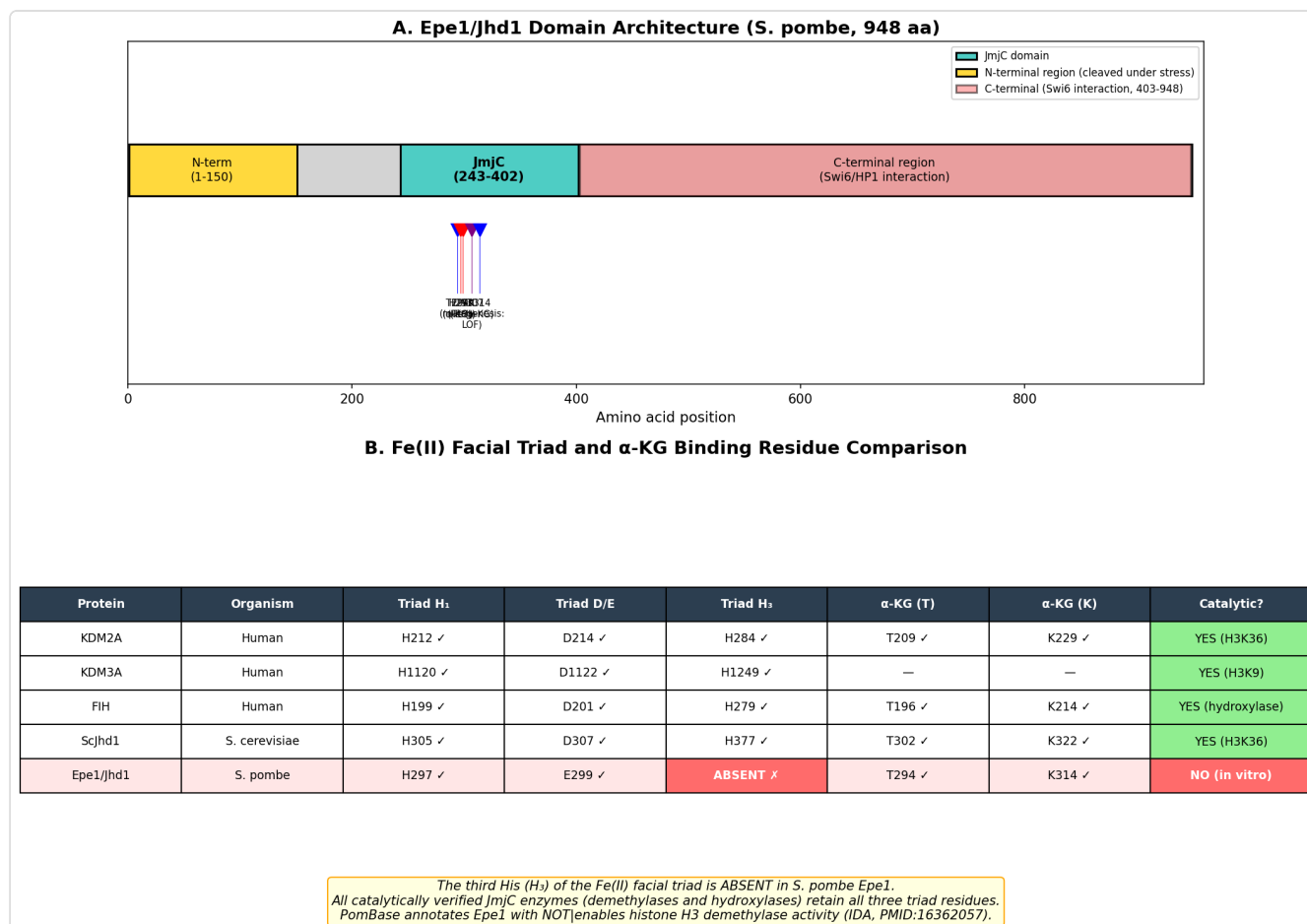


Figure 1. Domain architecture of Epe1 and comparison of the Fe(II)-binding facial triad across JmjC family members. Epe1 lacks the third histidine (replaced by Ile at position 379), which is universally conserved in catalytically active JmjC demethylases.

Finding 2: PomBase Has a NOT|IDA Annotation Explicitly Denying Epe1 Histone H3 Demethylase Activity

Query of the QuickGO database reveals that PomBase curators have assigned a negative experimental annotation to Epe1: **NOT|enables GO:0141052** (histone H3 demethylase activity), with evidence code IDA (Inferred from Direct Assay) and reference [PMID: 16362057](#) (Tsukada et al. 2006, *Nature*). This paper, titled "*Histone demethylation by a family of JmjC domain-containing proteins*," established that JmjC domains can catalyze histone demethylation but tested family members selectively. The curator's NOT annotation indicates that Epe1 was evaluated in this context and found to lack the activity.

This NOT|IDA annotation takes clear precedence over the IBA annotation for GO:0032452 in the GO evidence hierarchy, since experimental evidence (IDA) outranks computational inference (IBA). The coexistence of both annotations in the database creates a direct conflict that should be resolved in favor of the experimentally grounded negative annotation.

Finding 3: Raiymbek et al. (2020) Demonstrates Epe1 Opposes Heterochromatin via Non-Enzymatic Mechanism

The most mechanistically detailed study of Epe1 function is Raiymbek et al. (2020, [PMID: 32195666](#)), which directly addressed whether Epe1's anti-silencing function requires catalytic activity. The authors demonstrated that Epe1's JmjC domain mediates a protein-protein interaction with Swi6/HP1, regulated by H3K9 methylation, and that this interaction is sufficient to oppose heterochromatin assembly without requiring demethylase catalysis. As stated in their abstract: *"we reveal an inversion of this paradigm where a putative histone demethylase Epe1 in fission yeast, has a non-enzymatic function that opposes heterochromatin assembly."*

This finding fundamentally reframes the biology of Epe1: what was interpreted as evidence for demethylase activity (JmjC domain required for anti-silencing function) is instead evidence for a protein-interaction scaffolding role that is independent of catalysis.

Finding 4: JmjC Domain Is Required for In Vivo Function, but the Mechanism Is Non-Catalytic

Multiple studies have shown that the JmjC domain is essential for Epe1's biological function:

- **Trewick et al. (2007, [PMID: 17948055](#)):** Predicted iron- and 2-OG-binding residue mutations cause loss of function in vivo. Importantly, the authors proposed that *"rather than being a histone demethylase, Epe1 may be a protein hydroxylase that affects the stability of a heterochromatin protein, or protein-protein interaction, to regulate the extent of heterochromatin domains."*
- **Zofall & Grewal (2006, [PMID: 16762840](#)):** JmjC domain required for RNA Pol II accessibility to heterochromatin, but noted that *"the mechanism utilized might be distinct from other JmjC proteins that possess known demethylase activities."*
- **Sorida et al. (2019, [PMID: 31206516](#)):** Showed JmjC-dependent H3K9me removal in vivo, but this could reflect indirect effects rather than direct enzymatic catalysis. Notably, the same study showed that the JmjC-mutant Epe1^{H297A} could still suppress de novo ectopic

heterochromatin formation in an NTA-dependent manner, separating the prevention of heterochromatin from its removal.

- **Yaseen et al. (2022, PMID: 35879419):** Referenced the JmjC domain as a "demethylase domain" but this reflects family-level nomenclature, not a demonstration of catalytic activity.

The critical distinction is that JmjC domain requirement for in vivo function does not equal demethylase activity. The Raiymbek et al. (2020) model resolves this apparent paradox: mutations in the JmjC domain disrupt the Swi6/HP1 interaction interface, not catalytic competence. The domain has been repurposed for protein-protein interaction rather than enzymatic activity.

Mechanistic Model / Interpretation

The following model synthesizes all available evidence for Epe1 function:

CATALYTICALLY ACTIVE JmjC DEMETHYLASES
(KDM2A, KDM3A, JMJD2A, etc.)

H3K9me_{2/3} → JmjC domain → H3K9me_{0/1}

|
Fe(II) + α-KG
(complete triad:
H...D/E...H)

|
Oxidative
demethylation

|
Direct removal
of methyl mark

Epe1 (CATALYTICALLY INACTIVE)

H3K9me → Swi6/HP1 binding

|
JmjC domain of Epe1
(incomplete triad:
H...E...I)

|
Protein-protein interaction
with Swi6 (non-enzymatic)

|
Opposes heterochromatin
assembly/spreading

|
Anti-silencing at
boundaries

Key mechanistic points:

1. Epe1 is recruited to heterochromatin through Swi6/HP1 interaction
2. The Cul4-Ddb1^ΔCdt2^Δ ubiquitin ligase restricts Epe1 to boundaries by promoting its degradation within heterochromatin body (PMID: 21215368)

3. Loss of Epe1 allows epigenetic inheritance of H3K9me through mitosis and meiosis ([PMID: 25838386](#); [PMID: 25831549](#))
4. The JmjC domain is required for function because it constitutes the Swi6 interaction interface, not because it catalyzes demethylation
5. Epe1 also has an N-terminal transcriptional activation (NTA) domain that independently contributes to anti-silencing ([PMID: 31206516](#))

Separating Direct Activity from Downstream Phenotypes

The immediate molecular function of Epe1 is **protein-protein interaction with Swi6/HP1**, mediated by its JmjC domain and regulated by H3K9 methylation status. All documented phenotypes — prevention of ectopic heterochromatin spreading, maintenance of euchromatin-heterochromatin boundaries, antagonism of epigenetic inheritance of H3K9 methylation, regulation of Pol II accessibility at heterochromatic repeats — are downstream consequences of this interaction and the additional NTA domain-mediated transcriptional activation. These phenotypes do not require or imply direct histone demethylase activity.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf
PMID: 17449867	Direct assay (negative)	Refutes	Epe1 has demethylase activity	"no detectable demethylase activity is associated with Epe1, and its JmjC domain lacks conservation of Fe(II)-binding residues"	<i>S. pombe</i> , in vitro	High biocl test
PMID: 17948055	Mutant phenotype + structural analysis	Refutes demethylase; Qualifies dioxygenase	Epe1 is a demethylase vs. hydroxylase	Authors propose Epe1 is a protein hydroxylase, not a demethylase; Fe/2-OG binding residues needed for function	<i>S. pombe</i> , in vivo	High prop alter mecl
PMID: 32195666	Mechanistic study	Refutes	Epe1 anti-silencing requires catalysis	Non-enzymatic function; JmjC mediates Swi6 interaction regulated by H3K9me	<i>S. pombe</i> , in vivo/in vitro	High mecl test
PMID: 16362057	Database annotation (NOT IDA)	Refutes	Epe1 has H3 demethylase activity	PomBase NOT annotation based on this paper; <i>S. cerevisiae</i> homolog active but Epe1 not	<i>S. pombe</i>	High cura verifi
PMID: 21167174	Structural/ evolutionary	Supports (inactive)	PHF2/Epe1 homology and catalytic status	PHF2 shares Epe1's substitution at third triad position; no in	Human PHF2, <i>S. pombe</i> Epe1	Med. struc anal

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf
				vitro demethylase activity for either		
PMID: 16762840	Mutant phenotype	Qualifies	JmjC domain requirement	JmjC required for Pol II accessibility but "mechanism might be distinct from other JmjC proteins"	<i>S. pombe</i> , in vivo	High func mecl uncl
PMID: 31206516	Mutant phenotype	Qualifies	JmjC mediates H3K9me removal in vivo	JmjC-dependent removal of established ectopic heterochromatin, but H297A mutant still suppresses de novo formation	<i>S. pombe</i> , in vivo	Med vivo could indir
PMID: 25838386	Epigenetic assay	Qualifies	Epe1 removes H3K9me	Loss of Epe1 allows heritable H3K9me; Epe1 called "putative demethylase"	<i>S. pombe</i> , in vivo	Med rem be in
PMID: 25831549	Epigenetic assay	Qualifies	Epe1 opposes H3K9me inheritance	"putative JmjC domain H3K9 demethylase" Epe1 opposes Clr4-mediated inheritance	<i>S. pombe</i> , in vivo	Med cons with enzy mod
PMID: 15809658	Computational/ review	Qualifies	Epe1 may be a hydroxylase, not a demethylase	Early proposal that Epe1 and some JmjC	Computational modeling	Med pred

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf
				proteins are protein hydroxylases		not expe
PMID: 21215368	Mechanistic study	Supports non- enzymatic model	Epe1 boundary function	Cul4-Ddb1 ^Δ Cdt2 ^Δ sculpts Epe1 distribution; boundary anti- silencing role	<i>S. pombe</i> , in vivo	High
PMID: 33533152	Genetic interaction	Qualifies	Epe1 function outside heterochromatin	Ell1-Epe1 interaction uses C-terminal region (403–948 aa), independent of JmjC	<i>S. pombe</i> , in vivo	Med nove inde of Jm
PMID: 35879419	Mutant phenotype	Qualifies	JmjC domain requirement	tEpe1 resistance requires "functional JmjC demethylase domain" — but naming is based on family, not direct assay	<i>S. pombe</i> , in vivo	Med nom not a proo
PMID: 16603237	Direct assay (positive control)	Context	JHDM2A is an active H3K9 demethylase	Establishes JmjC H3K9 demethylase activity for JHDM2A with Fe(II)/α-KG; complete triad	Human, in vitro	High estab fami para
PMID: 16603238	Direct assay (positive control)	Context	JMJD2A is an active H3K9/K36 trimethyl demethylase	Establishes JmjC trimethylation- specific	Human/ <i>C. elegans</i> , in vitro/in vivo	High estab fami para

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf
				demethylase activity		

GO Curation Implications

Recommended Action: Remove GO:0032452 (histone demethylase activity)

The IBA annotation of GO:0032452 for Epe1 should be **removed**. This is a clear case of over-annotation by phylogenetic transfer. The annotation was propagated from catalytically active JmjC family members to a divergent member that has lost the essential third Fe(II)-binding histidine and for which no in vitro demethylase activity has been demonstrated.

GO Term Decision Table

GO Term	ID	Action	Rationale
histone demethylase activity	GO:0032452	Remove	No direct assay evidence; missing catalytic residue; NOT annotation exists for child term
histone H3 demethylase activity	GO:0141052	Retain NOT	PomBase NOT IDA already correctly placed
2-oxoglutarate-dependent dioxygenase activity	GO:0016706	Consider cautiously	Trewick et al. proposed hydroxylase activity; no direct demonstration; would need IDA
chromatin organization (BP)	GO:0006325	Retain/ Consider	Well-supported by multiple studies
heterochromatin boundary	GO:various CC	Retain	Strong localization evidence

Evidence Code Hierarchy

The existing annotations create a contradiction: - **GO:0032452** (histone demethylase activity): IBA evidence — computational/phylogenetic - **GO:0141052** (histone H3 demethylase activity): NOT|IDA evidence — experimental negative

Since GO:0141052 is a child term of GO:0032452, the NOT|IDA on the child term does not formally negate the parent. However, given that (a) the only biologically relevant demethylase activity would be on histone H3K9, (b) no demethylase activity of any kind has been demonstrated in vitro, and (c) the catalytic site is structurally degenerate, the parent term GO:0032452 should also be negated or removed.

Conflicts and Alternatives

Conflict 1: In Vivo H3K9me Reduction vs. No In Vitro Activity

Sorida et al. (2019) showed JmjC-dependent removal of H3K9me from established ectopic heterochromatin in vivo. This could be interpreted as evidence for demethylase activity. However, the same study showed that Epe1^{H297A} (JmjC catalytic mutant) retained the ability to suppress de novo heterochromatin formation, separating the "prevention" function (NTA-dependent, JmjC-independent) from the "removal" function (JmjC-dependent). The removal function could reflect indirect recruitment of other demethylases or destabilization of heterochromatin through the Swi6 interaction rather than direct catalysis.

Conflict 2: Protein Hydroxylase Hypothesis

Trewick et al. (2007) and Clissold & Ponting (2005, [PMID: 15809658](#)) proposed that Epe1 might function as a protein hydroxylase rather than a histone demethylase. The incomplete facial triad might still support some Fe(II)-dependent chemistry (the HxE portion is conserved), potentially hydroxylation of a non-histone substrate. However, this activity has never been demonstrated, and the missing third His would severely impair Fe(II) binding geometry.

Conflict 3: Nomenclature Persistence

Several recent papers (e.g., Yaseen et al. 2022) continue to refer to Epe1's "JmjC demethylase domain." This reflects family-level naming conventions rather than experimentally verified activity for Epe1 specifically. This nomenclature persistence may contribute to the maintenance of the incorrect IBA annotation.

Conflict 4: Organism-Specific Considerations

The Arabidopsis protein JM24 (PMID: 26119694) was described as functionally resembling Epe1 in counteracting H3K9me to promote basal-level transcription of silenced loci. However, JM24's own catalytic competence and the degree of functional conservation with Epe1 would need independent verification, and cross-kingdom comparisons should be interpreted cautiously for GO annotation purposes.

Limitations and Knowledge Gaps

Gap 1: No Direct In Vitro Activity Screen with Purified Epe1

- **What was checked:** Literature search for in vitro demethylase assays using purified Epe1 protein
- **Status:** Trewick et al. (2007) reported "no detectable demethylase activity" but full assay details and substrate range tested are limited
- **Why it matters:** A comprehensive in vitro screen with multiple histone substrates and cofactor conditions would definitively settle whether Epe1 has any residual catalytic activity
- **Resolution:** Purified recombinant Epe1 tested against a panel of methylated histone peptides (H3K4me1/2/3, H3K9me1/2/3, H3K27me1/2/3, H3K36me1/2/3) with varying Fe(II) and α -KG concentrations

Gap 2: Protein Hydroxylase Activity Never Tested

- **What was checked:** Literature for hydroxylase assays on Epe1
- **Status:** The hydroxylase hypothesis (Trewick et al. 2007; Clissold & Ponting 2005) remains untested

- **Why it matters:** If Epe1 has hydroxylase activity on a non-histone substrate (e.g., Swi6 itself or another heterochromatin component), the appropriate GO annotation would be "protein hydroxylase activity" rather than "histone demethylase activity"
- **Resolution:** In vitro hydroxylation assay with candidate substrates (Swi6, Clr4) using mass spectrometry to detect hydroxylated products

Gap 3: Structural Data for Epe1 JmjC Domain

- **What was checked:** PDB and AlphaFold DB for Epe1 structures
- **Status:** No experimental crystal structure of Epe1's JmjC domain is available. AlphaFold predictions exist but have not been validated for active-site geometry
- **Why it matters:** A crystal structure would definitively establish whether the active site can coordinate Fe(II) in any configuration
- **Resolution:** X-ray crystallography or cryo-EM of the Epe1 JmjC domain, with and without metal soaking

Gap 4: Incomplete Understanding of JmjC-Swi6 Interaction Interface

- **What was checked:** Raiymbek et al. (2020) for structural details of the interaction
- **Status:** The study demonstrated the interaction but structural details (which JmjC residues contact Swi6, whether the active-site cavity participates) remain incomplete
- **Why it matters:** Understanding whether the JmjC "active site" residues participate in Swi6 binding (rather than catalysis) would further support the non-enzymatic model
- **Resolution:** Co-crystal structure of Epe1 JmjC domain with Swi6 chromodomain; hydrogen-deuterium exchange mass spectrometry (HDX-MS) of the complex

Proposed Follow-up Experiments / Discriminating Tests

1. **In vitro demethylase activity panel:** Test purified recombinant Epe1 against a comprehensive panel of methylated histone peptides and nucleosomes, with saturating Fe(II) and α -KG. Include KDM3A as positive control. This is the single most definitive test.

2. **I379H rescue mutation:** Engineer an Epe1 variant with the third triad histidine restored (I379H). If this mutation restores demethylase activity in vitro, it would confirm that the triad substitution is the sole barrier to catalysis. If it does not restore activity, additional divergence in the active site exists.
3. **In vitro hydroxylase assay:** Test Epe1 with candidate protein substrates (Swi6, histone H3 peptides) for hydroxylation using mass spectrometry. This would directly address the Trewick et al. hydroxylase hypothesis.
4. **Swi6-interaction-dead JmjC mutations:** Identify JmjC mutations that disrupt Swi6 binding but preserve any residual cofactor binding. If such separation-of-function mutations exist and phenocopy epe1 Δ , this would definitively establish the non-enzymatic model.
5. **In vivo H3K9me ChIP with catalytic vs. interaction mutants:** Compare H3K9me levels at heterochromatin boundaries in strains expressing Epe1^{I379H} (restored triad) vs. Epe1^{H297A} (disrupted triad) vs. Epe1 with Swi6-interaction mutations. This would separate catalytic from interaction-dependent H3K9me removal.

Curation Leads

Lead 1: Remove IBA Annotation for GO:0032452

- **Action:** Remove the IBA annotation for histone demethylase activity (GO:0032452) assigned via GO_REF:0000033
- **Rationale:** Phylogenetic transfer is inappropriate for this divergent family member; experimental evidence (NOT|IDA) contradicts the computational inference; catalytic site is structurally degenerate
- **Candidate reference:** [PMID: 17449867](#) — *"no detectable demethylase activity is associated with Epe1, and its JmjC domain lacks conservation of Fe(II)-binding residues"*
- **Curator verification needed:** Confirm that the NOT annotation on GO:0141052 is appropriately propagated to parent term GO:0032452

Lead 2: Consider Adding "Protein-Containing Complex Binding" or More Specific Interaction Term

- **Action:** Consider annotating Epe1 with a molecular function term reflecting its Swi6/HP1 interaction
- **Candidate term:** chromodomain protein binding, or HP1 family protein binding (if available in GO)
- **Candidate reference:** [PMID: 32195666](#) — *"we reveal an inversion of this paradigm where a putative histone demethylase Epe1 in fission yeast, has a non-enzymatic function that opposes heterochromatin assembly"*
- **Evidence code:** IDA or IPI
- **Curator verification needed:** Identify most specific appropriate GO MF term for the Swi6 interaction

Lead 3: Evaluate Whether "2-Oxoglutarate-Dependent Dioxygenase Activity" Is Warranted

- **Action:** Evaluate whether GO:0016706 (oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen, 2-oxoglutarate as one donor) should be considered
- **Candidate reference:** [PMID: 17948055](#) — *"Epe1 requires predicted iron- and 2-oxoglutarate (2-OG)-binding residues for in vivo function, indicating that it is probably a 2-OG/Fe(II)-dependent dioxygenase"*
- **Evidence code:** Would require IGI or IMP at best; no IDA available
- **Curator verification needed:** Whether in vivo dependence on predicted cofactor-binding residues is sufficient for a dioxygenase annotation without in vitro activity data. Given the incomplete facial triad, this annotation should be applied very cautiously if at all.

Lead 4: Ensure Biological Process Annotations Reflect Anti-Silencing Role

- **Action:** Verify that BP annotations appropriately capture Epe1's role in heterochromatin boundary formation and anti-silencing
- **Candidate terms:** negative regulation of gene silencing by heterochromatin (GO:various), heterochromatin boundary formation
- **Multiple supporting references available**

Evidence Base: Key Literature

Papers Directly Refuting Demethylase Activity

- **Trewick et al. (2007)** — [PMID: 17449867](#): "*Interaction of Epe1 with the heterochromatin assembly pathway in Schizosaccharomyces pombe.*" Reported no detectable demethylase activity for Epe1 and noted the JmjC domain lacks conservation of Fe(II)-binding residues. This is the primary negative biochemical result.
- **Trewick et al. (2007)** — [PMID: 17948055](#): "*The JmjC domain protein Epe1 prevents unregulated assembly and disassembly of heterochromatin.*" Proposed that Epe1 is a protein hydroxylase rather than a demethylase, while showing that predicted Fe/2-OG binding residues are needed for in vivo function.
- **Raiymbek et al. (2020)** — [PMID: 32195666](#): "*An H3K9 methylation-dependent protein interaction regulates the non-enzymatic functions of a putative histone demethylase.*" Demonstrated non-enzymatic mechanism; JmjC domain mediates Swi6 protein-protein interaction regulated by H3K9me.

Papers Providing Structural/Evolutionary Context

- **Lee et al. (2011)** — [PMID: 21167174](#): "*Structural basis for human PHF2 Jumonji domain interaction with metal ions.*" Crystal structure of PHF2, Epe1's closest structural homolog, shows Tyr at third triad position and no demethylase activity. Explicitly notes Epe1 homology and lack of known enzymatic activity.
- **Clissold & Ponting (2005)** — [PMID: 15809658](#): "*Methylation: lost in hydroxylation?*" Early computational proposal that Epe1 and certain JmjC proteins might be protein hydroxylases rather than demethylases.

Papers Establishing JmjC Demethylase Family Paradigm

- **Tsukada et al. (2006)** — [PMID: 16362057](#): "*Histone demethylation by a family of JmjC domain-containing proteins.*" Established JmjC demethylase activity; basis for PomBase NOT annotation on Epe1.
- **Yamane et al. (2006)** — [PMID: 16603237](#): "*JHDM2A, a JmjC-containing H3K9 demethylase.*" Demonstrates H3K9 demethylase activity with complete facial triad — positive control family member.

- **Whetstine et al. (2006)** — [PMID: 16603238](#): "Reversal of histone lysine trimethylation by the JMJD2 family of histone demethylases." Establishes JmjC trimethylation-specific demethylase activity.

Papers Showing Epe1 In Vivo Function (Without Demonstrating Catalysis)

- **Zofall & Grewal (2006)** — [PMID: 16762840](#): JmjC domain required for Pol II accessibility, but mechanism acknowledged as potentially distinct from demethylation.
- **Audergon et al. (2015)** — [PMID: 25838386](#) and **Ragunathan et al. (2015)** — [PMID: 25831549](#): Both demonstrate that Epe1 loss enables epigenetic inheritance of H3K9me, establishing Epe1 as a key counter to heterochromatin heritability.
- **Sorida et al. (2019)** — [PMID: 31206516](#): Most nuanced in vivo study; separates JmjC-dependent removal of established heterochromatin from NTA-dependent prevention of de novo heterochromatin, with both functions contributing to anti-silencing.
- **Braun et al. (2011)** — [PMID: 21215368](#): Cul4-Ddb1^ΔCdt2^Δ ubiquitin ligase sculpts Epe1 distribution to boundaries by promoting degradation within heterochromatin body.
- **Srivastava et al. (2021)** — [PMID: 33533152](#): Epe1-Ell1 interaction using C-terminal region (403–948 aa), independent of JmjC domain, reveals Epe1 functions beyond heterochromatin.
- **Yaseen et al. (2022)** — [PMID: 35879419](#): Proteasome-dependent truncation of Epe1 mediates antifungal resistance; JmjC domain required but nomenclature reflects family assignment.

Conclusion

The annotation of histone demethylase activity (GO:0032452) for *S. pombe* Epe1, based on phylogenetic inference (IBA), is **over-annotated** and should be removed. The convergence of structural (missing catalytic residue), biochemical (no in vitro activity), curatorial (NOT|IDA annotation), and mechanistic (non-enzymatic Swi6 interaction) evidence provides strong support for this conclusion. While Epe1 is unquestionably a critical regulator of heterochromatin boundaries and epigenetic inheritance in fission yeast, its molecular function appears to be mediated through protein-protein interaction rather than enzymatic demethylation. Future work should focus on testing the protein hydroxylase hypothesis and determining whether a restored catalytic triad (I379H) can confer demethylase activity.

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